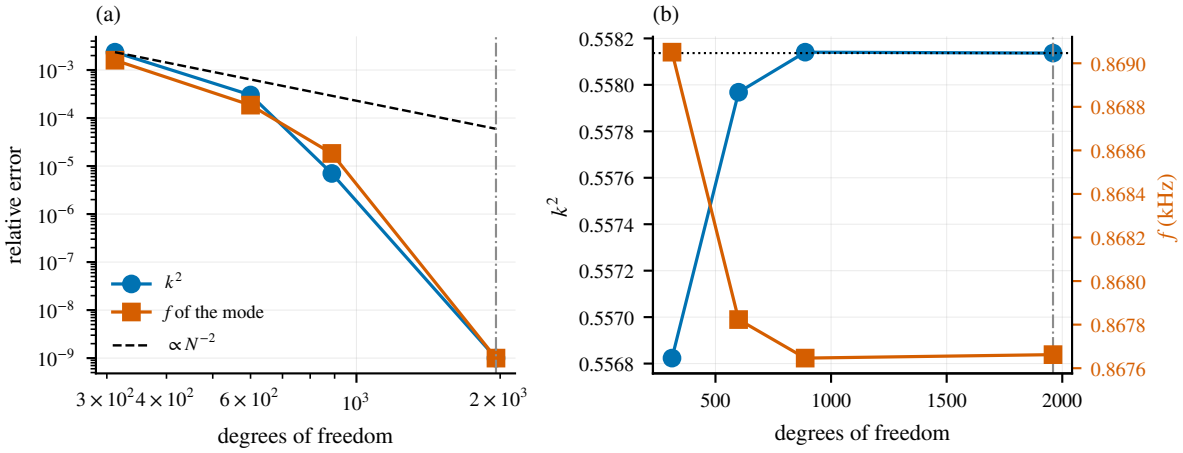
(a)
relative error
10^{-3}
10^{-4}
10^{-5}
10^{-6}
10^{-7}
10^{-8}
10^{-9}
k^2
f of the mode
$\propto N^{-2}$
3×10^2 4×10^2
6×10^2
10^3
2×10^3
degrees of freedom
(b)
k^2
0.5582
0.5580
0.5578
0.5576
0.5574
0.5572
0.5570
0.5568
f (kHz)
0.8690
0.8688
0.8686
0.8684
0.8682
0.8680
0.8678
0.8676
500
1000
1500
2000
degrees of freedom